

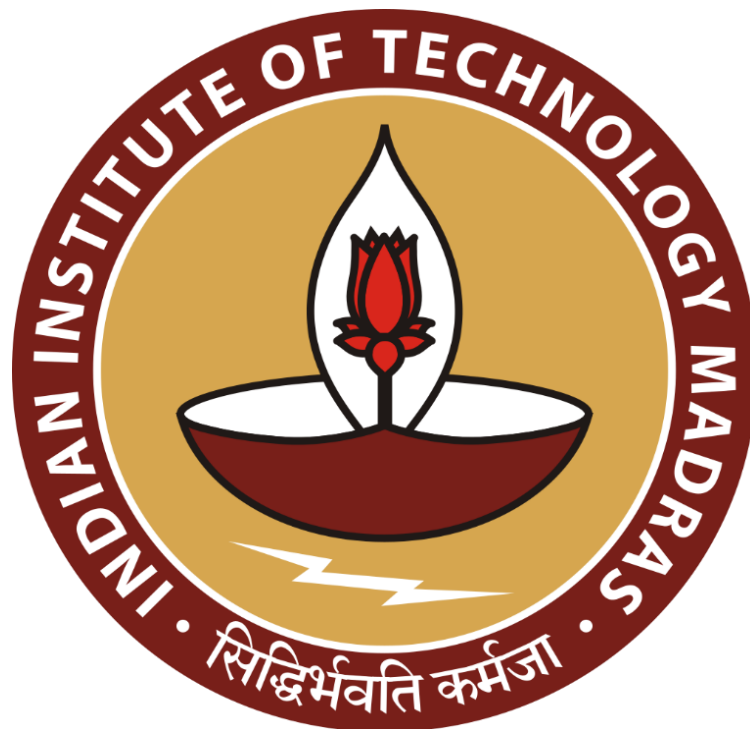
Overcoming Operational Bottlenecks In Laundry Services: A Case Of Uclean

A FINAL REPORT FOR THE BDM CAPSTONE PROJECT

Submitted By

Name: Aditi Saxena

Roll number: 23f1000290



IIT MADRAS BS DEGREE PROGRAM
Indian Institute of Technology Madras, Chennai
Tamil Nadu, India, 600036

Contents:

1. Executive Summary	2
2. Detailed Explanation of Analysis Process	3-8
3. Results and Findings	8-14
4. Interpretations of Results and Recommendations	14-18

1. EXECUTIVE SUMMARY

This Capstone Project on Business Data Management (BDM) analyzes operational performance at UClean's Sector 37D, Gurgaon franchise, with the objective of improving service efficiency, reducing cancellations, and strengthening seasonal demand forecasting. By examining turnaround time patterns, cancellation trends, staff- wise workload distribution, and monthly demand fluctuations, the study provides data- driven insights to enhance operational planning and overall business sustainability.

The report begins with data preprocessing, which involved standardizing date formats, cleaning pricing variables (₹), categorizing service types, and computing Turnaround Time using the formula $Delay\ (Days) = Actual\ Delivery\ Date - Expected\ Delivery\ Date$.

Exploratory Data Analysis (EDA) was conducted using Pivot Tables and visualizations such as bar charts and line graphs to evaluate delay frequency, cancellation patterns, service- wise revenue distribution, and seasonal demand variation.

Key findings indicate that moderate delays of 2-3 days occurred most frequently, particularly during peak- demand months (April-May), reflecting workload imbalance and capacity constraints. Cancellation rates were significantly higher in these months, primarily due to late pickups and staff shortages. Seasonal analysis revealed a demand surge in early months followed by a decline in June-July, highlighting reactive rather than proactive resource management.

To improve performance, demand- based workforce scheduling, systematic delay monitoring, and forecasting- based capacity planning should be implemented. Aligning staffing and service prioritization with seasonal demand patterns can reduce operational strain, minimize revenue loss, and improve customer satisfaction.

A data- driven operational strategy will enable UClean to streamline workflows, optimize resource allocation, and achieve sustainable service excellence in a competitive B2C environment.

2. DETAILED EXPLANATION OF ANALYSIS PROCESS

2.1 Analysis Overview

The aim of this analysis is to evaluate the operational efficiency of UClean's Sector 37D, Gurgaon franchise, with emphasis on reducing turnaround delays, minimizing order cancellations, and managing seasonal demand fluctuations. The study examines order timelines, staff workload distribution, service-wise demand patterns, cancellation behavior, and monthly demand variation to identify operational bottlenecks affecting service reliability and business performance. Particular attention is given to recurring delays, high cancellation rates during peak demand periods, and uneven workforce allocation contributing to operational inefficiencies.

Kindly refer to the analysis folder for details: [BDM Project](#)

A structured data-driven methodology was adopted, beginning with data acquisition, preprocessing, exploratory data analysis, statistical evaluation, and seasonal demand comparison. Techniques such as delay frequency analysis, cancellation rate computation, staff workload assessment, and demand comparison across months were applied to ensure systematic evaluation of operational performance. The insights derived from this analysis support demand-based workforce allocation, improved pickup scheduling, and proactive capacity planning to reduce service delays, minimize cancellations, and enhance overall operational efficiency.

2.2 Data Acquisition and Processing

The dataset used for this analysis was compiled in a structured Excel workbook titled "UClean Laundry Data", consisting of **1,116 order** covering the period from **April 2025 to July 2025**. The dataset represents operational records of a laundry service business and includes information related to order timelines, service types, revenue, cancellations, and staff allocation. The data was structured at order level, enabling detailed analysis of operational performance across time, workforce distribution, and service categories.

To refine the dataset for meaningful analysis, several preprocessing steps were performed to ensure consistency, accuracy, and reliability of results.

- **Data Cleaning** - Currency symbols (₹) and formatting inconsistencies were removed from

revenue fields to convert them into numeric format. Missing values and inconsistent entries were identified and corrected to maintain data integrity.

- **Date Standardization** - All date columns were formatted uniformly using a consistent date structure to enable accurate calculation of turnaround time and monthly aggregation.

- **Turnaround Time Calculation** - A derived variable, Delay (Days), was computed using the formula:

$$\text{Delay (Days)} = \text{Actual Delivery Date} - \text{Expected Delivery Date}.$$

This variable measures turnaround inefficiency and forms the basis for delay distribution analysis.

- **Cancellation Encoding** - Cancelled orders were identified using the Status field. Missing delivery dates for cancelled orders were retained to differentiate operational failures from completed services.

- **Duplicate Removal** - Duplicate order entries were identified and removed to prevent skewed statistical results and ensure accurate calculations.

- **Data Consistency Checks** - Service categories and staff identifiers were standardized to maintain uniformity across the dataset and ensure reliable analysis.

These preprocessing steps ensured that the dataset remained consistent, structured, and suitable for statistical evaluation, enabling accurate identification of operational inefficiencies and demand patterns.

2.3 Exploratory Data Analysis and Trend :

Following data cleaning and preprocessing, Exploratory Data Analysis (EDA) was conducted to examine operational inefficiencies, delay patterns, cancellation trends, staff workload distribution, and seasonal demand fluctuations at UClean. The objective of this analysis was to identify key operational bottlenecks affecting turnaround time, service reliability, and demand- driven workload variation.

- **Turnaround Time and Delay Distribution Analysis** - Delay frequency was analyzed by grouping orders into categories ranging from 0 to 5 days. This helped identify the most common delay range and recurring operational inefficiencies. Employee- wise delay distribution was also examined to detect workload imbalance and determine whether delays were concentrated among specific staff members.

- **Staff Workload Assessment** - Order allocation across staff members was evaluated using staff- wise order counts and delay distribution. This analysis helped identify overloaded staff handling a higher number of delayed orders and underutilized staff with lower workload, indicating inefficient workforce allocation.
- **Cancellation Pattern Analysis** - Cancellation rate was calculated using the ratio of cancelled orders to total orders. Monthly cancellation trends were analyzed along with cancellation reasons such as late pickup, staff shortage, and inventory shortage to identify dominant operational failure points and quantify service inefficiencies.
- **Seasonal Demand Trend Analysis** - Monthly aggregation of total orders and revenue was performed to identify demand spikes and off- season decline. This helped determine how demand fluctuations influenced delays, cancellations, and operational workload.
- **Service wise Demand Analysis** - Orders were analyzed across service categories such as Dry Clean, Wash & Fold, Premium Laundry, Steam Iron, and Wash & Iron. This analysis helped identify high- demand services and evaluate workload distribution across service types.

Visual tools such as bar charts, grouped bar charts, and line graphs were used to represent demand trends, delay distribution, and cancellation patterns. These visualizations helped identify patterns, highlight operational bottlenecks, and support data- driven decision making.

2.4 Statistical Summary and Quantitative Trends

The dataset was initially examined to ensure accuracy and consistency before analysis. This involved validating column names, checking data types, identifying missing values, and resolving duplicate records to maintain data integrity. Calculated fields such as delay days, cancellation indicators, and monthly aggregation values were verified to ensure reliable statistical evaluation.

Essential statistical measures such as mean, median and frequency counts were computed to summarize operational performance across different time periods. The statistical summary helped quantify delay distribution, measure cancellation percentage, evaluate monthly demand variation, and identify workload imbalance affecting operational efficiency.

- **Turnaround Time and Delay Analysis** - Delay days were summarized using mean and

median values to measure service efficiency. Frequency distribution of delays ranging from 0 to 5 days was computed to identify the most common delay range. This analysis highlighted recurring moderate delays and confirmed that turnaround inefficiencies increased during peak demand months.

- **Cancellation Rate and Service Failure Analysis** - Cancellation percentage was calculated month wise to evaluate operational reliability. Total cancelled orders and cancellation reasons were summarized to identify dominant operational bottlenecks. This helped quantify service failures and assess the impact of workforce limitations and pickup delays.

- **Seasonal Demand and Order Volume Analysis** - Monthly aggregation of total orders was performed to detect demand variation across April-July. Statistical comparison between peak demand months and off season months helped measure workload concentration and operational pressure. This analysis confirmed that demand fluctuation significantly influenced delays and cancellations.

2.5 Seasonal Demand Comparison Analysis

To analyze seasonal demand fluctuation, monthly order volume was aggregated and compared across April to July. The dataset was divided into peak demand months (April-May) and off season months (June-July) to evaluate variations in order volume, revenue, and operational workload. This comparison helped identify demand concentration during peak periods and reduced service pressure during low demand months. The analysis also examined service wise demand distribution to determine which services contributed most to workload during peak periods.

This method was selected because seasonal comparison directly links demand variation with operational performance. By comparing peak and off season periods, the analysis quantifies workload imbalance and evaluates its impact on turnaround delays and cancellations. These insights help identify capacity limitations and support demand based workforce allocation and improved scheduling to enhance operational efficiency.

2.6 Visual Representation Techniques:

Visual analytics were used to support interpretation of statistical findings and highlight operational inefficiencies. Graphical representations help convert numerical data into actionable business insights and allow comparison across multiple operational variables.

The following visualization techniques were used:

- Bar Charts - Used for monthly order volume and revenue comparison to identify peak demand periods
- Line Charts - Used to analyze demand trends and revenue variation across months
- Clustered Bar Charts - Used to compare cancellation reasons and service wise demand distribution
- Grouped Bar Charts - Used to analyze staff workload and delay relationship
- Comparative Charts - Used to evaluate seasonal demand variation across service types

All charts include labeled axes, units, and legends such as number of orders, delay days, and revenue (₹) to ensure clarity. These visualizations were selected because they effectively highlight operational bottlenecks related to delays, cancellations, and seasonal demand fluctuations.

2.7 Major Operational Challenges Identified

Based on exploratory analysis, statistical summary, and visual evaluation, key operational challenges affecting service efficiency were identified.

- High Turnaround Delays - Moderate delays of 2-3 days occurred most frequently, indicating recurring inefficiencies in processing and delivery timelines.
- High Cancellation Rate - A significant proportion of orders were cancelled, primarily due to late pickup and staff shortage, indicating operational capacity limitations.
- Seasonal Demand Fluctuation - Demand increased significantly during April-May and declined in June-July, creating workload imbalance and inconsistent operational performance.
- Uneven Staff Workload - Some staff handled significantly higher number of orders, resulting in increased delay frequency and service backlog.
- Capacity Constraint During Peak Demand - Operational resources remained constant despite demand variation, leading to delays and cancellations during peak months.

These challenges confirm that operational inefficiencies are demand driven and directly linked to workforce allocation, pickup scheduling, and seasonal demand variation. The

analysis establishes a quantitative foundation for interpretation and recommendation in the following section.

3. RESULTS AND FINDINGS :

All findings are presented in three sub-sections, each corresponding to one of the three problem statements. Each sub-section includes graphical representations followed by data-supported key findings.

3.1 Problem Statement : Long Turnaround Time & Service Delays

3.1.1 Delivery Delay Distribution

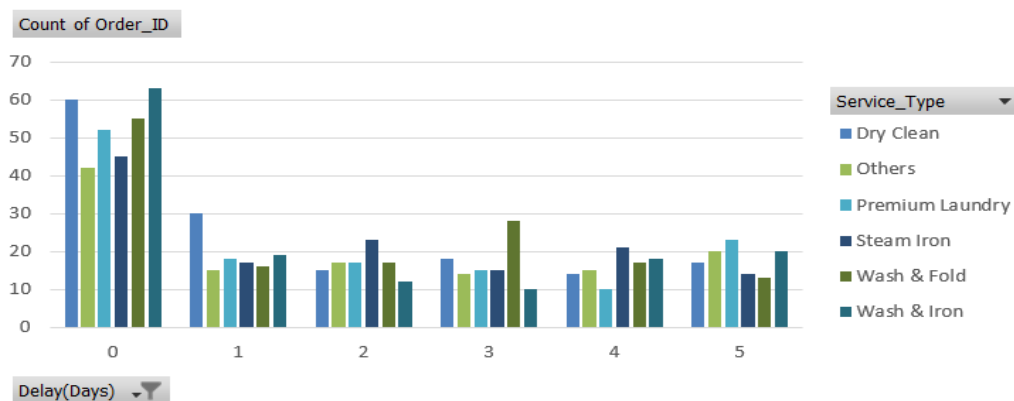


Figure 1: Delivery Delay Distribution -Completed Orders (n = 835)

The chart illustrates delay distribution across completed orders to evaluate turnaround efficiency. Out of 835 completed orders, only 317 orders were delivered on time (37.9%), while 518 orders experienced delays (62.1%). The most frequent delay range was 2-3 days, with 101 orders delayed by 2 days and 100 orders delayed by 3 days. Severe delays of 4-5 days accounted for 202 orders (24.2%). The average delay was calculated as 1.83 days, confirming consistent service delay.

Key Findings:

- Low On Time Delivery Rate - Only 317 out of 835 orders were delivered on time, representing 37.9% service efficiency.
- Majority Orders Delayed - 518 orders (62.1%) experienced delays, indicating recurring turnaround time inefficiencies.

- Moderate Delays Most Common - 2 day delays (101 orders) and 3 day delays (100 orders) together accounted for 24.1% of completed orders.
- Severe Delays Observed - 202 orders experienced 4-5 day delays, representing 24.2% and indicating service backlog.
- Average Delay Duration - Mean delay of 1.83 days confirms that delivery frequently exceeded expected turnaround time.

3.1.2 Staff Workload and Delay Relationship

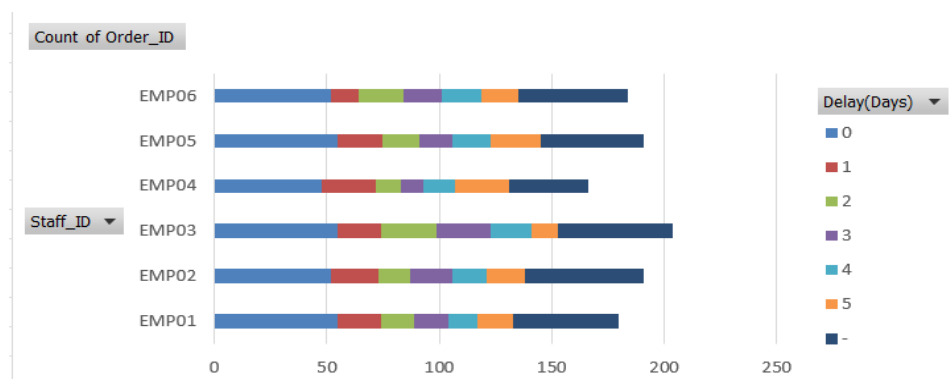


Figure 2: Staff-wise Delay Distribution -Order Volume vs. Delay Frequency per Staff Member

The chart shows staff wise delay distribution, highlighting workload imbalance. Some staff handled more than 200 orders, while others handled fewer, indicating uneven allocation. Staff with higher workload recorded higher delays across 2-5 day range. During peak demand, overloaded staff contributed to backlog while underutilized staff handled fewer orders.

Key Findings:

- Uneven Workload Distribution - Staff order handling ranged between 166 and 204 orders, indicating imbalance in allocation.
- Overloaded Staff Delays - Staff handling over 190 orders recorded higher delay frequency across 2-5 day range.
- Underutilized Capacity - Some staff handled significantly fewer orders, indicating inefficient workforce utilization.
- Peak Month Imbalance - Workload increased during April-May, causing backlog and delayed service delivery.
- Workload Delay Relationship - Higher order allocation directly resulted in increased delay frequency.

3.2 Problem Statement : High Order Cancellations & Operational Disruptions

3.2.1 Monthly Cancellation Rate Analysis

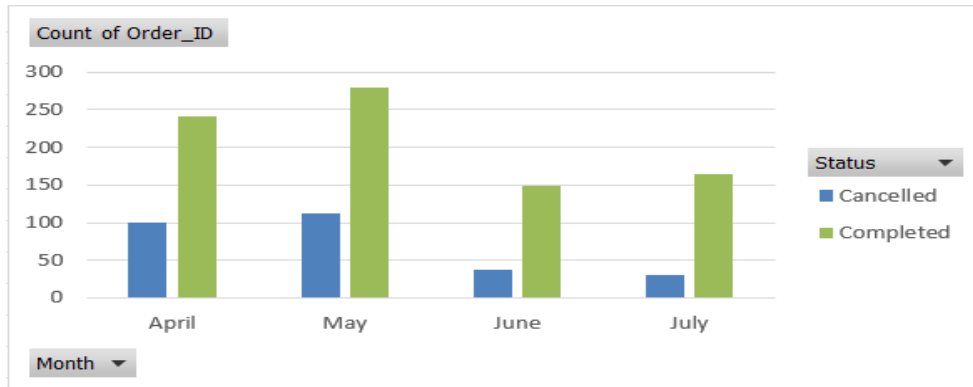


Figure 3: Month-wise Order Cancellation Distribution (April-July 2025)

The chart presents month wise cancellation distribution across April-July. Out of 1,116 total orders, 281 orders were cancelled, resulting in a cancellation rate of 25.18%. April recorded 100 cancellations (29.24%), while May recorded 113 cancellations (28.75%). In contrast, June recorded 38 cancellations (20.32%), and July recorded 30 cancellations (15.46%). The decline in cancellation rate during off season indicates capacity constraints during peak demand.

Key Findings:

- High Overall Cancellation Rate - 281 out of 1,116 orders were cancelled, resulting in 25.18% cancellation rate.
- Peak Month Cancellations - April recorded 100 cancellations (29.24%) and May recorded 113 cancellations (28.75%).
- Reduced Off- Season Cancellations - June recorded 38 cancellations (20.32%) and July recorded lowest at 15.46%.
- Demand Driven Failures - Higher order volume in peak months resulted in increased cancellations.
- Operational Capacity Limitation - Decline in cancellations during low- demand months indicates workforce constraints.

3.2.2 Cancellation Reason Breakdown

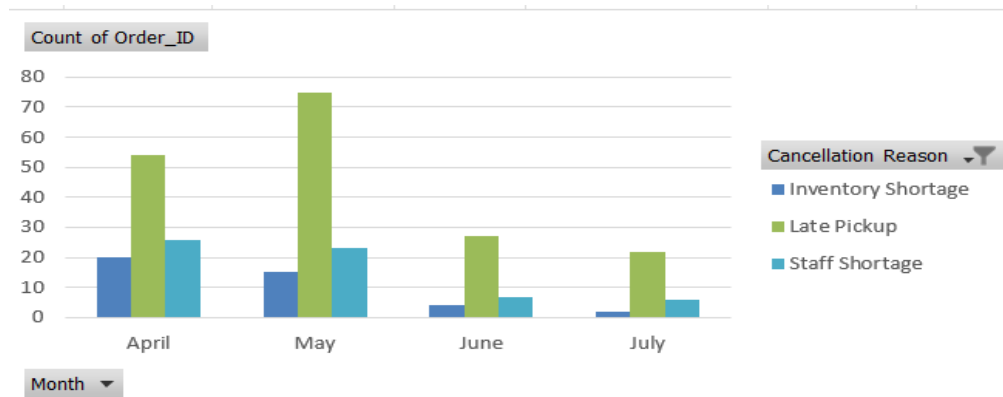


Figure 4: Cancellation Reason Distribution -281 Cancelled Orders

The chart illustrates distribution of cancellation reasons. Late pickup accounted for 178 cancellations (63.3%), making it the dominant cause. Staff shortage contributed 62 cancellations (22.1%), while inventory shortage accounted for 41 cancellations (14.6%). The dominance of pickup- related cancellations indicates inefficiencies in scheduling and workforce allocation.

Key Findings:

- Late Pickup Dominant Cause - 178 out of 281 cancellations (63.3%) occurred due to delayed pickup scheduling, making it the primary operational bottleneck.
- Staff Shortage Impact - 62 cancellations (22.1%) were caused by insufficient workforce, particularly during peak demand months.
- Inventory Constraints - 41 cancellations (14.6%) occurred due to resource unavailability, indicating planning inefficiencies.
- Pickup Stage Bottleneck - Majority cancellations occurred before processing, showing inefficiencies in initial service stage.
- Scheduling Inefficiency - High pickup- related cancellations indicate poor routing and demand- based workforce allocation.

3.3 Problem Statement : Seasonal Demand Fluctuation

3.3.1 Monthly Order Volume and Revenue Trend

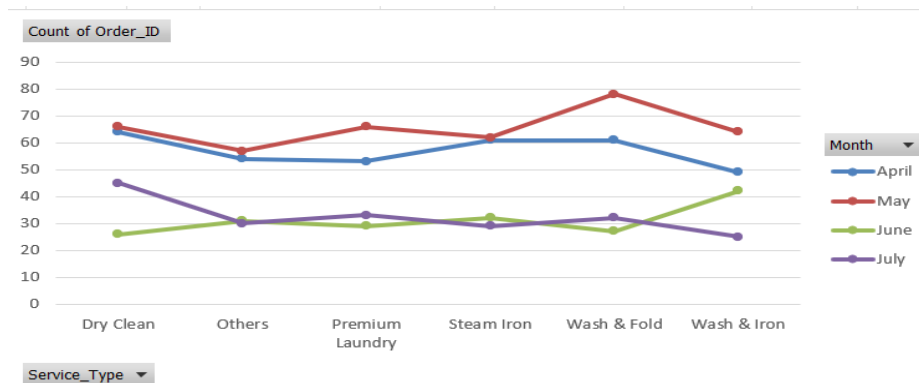


Figure 5: Month-wise Order Volume (April-July 2025)

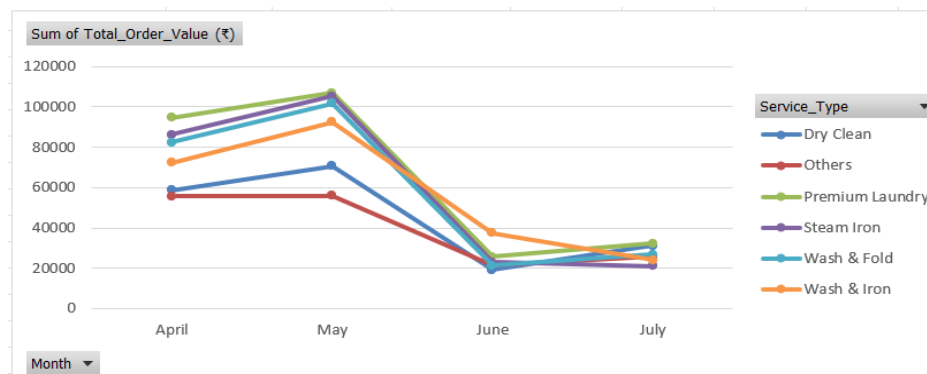


Figure 6: Month-wise Revenue Distribution (April-July 2025)

The charts illustrate month-wise variation in total orders and revenue from April to July 2025, highlighting strong seasonal demand fluctuation and its direct impact on financial performance. Order volume increased from 342 orders in April to 393 orders in May, representing a 14.9% growth and peak operational workload. However, demand declined sharply to 187 orders in June, showing a significant 52.4% drop compared to May. In July, demand slightly recovered to 194 orders, indicating only a marginal 3.7% increase and continued low demand. This uneven distribution confirms that operational workload is concentrated in peak months, where April and May together contributed 735 orders (65.8%), while June and July accounted for only 381 orders (34.2%).

Revenue followed a similar trend, confirming strong dependency on demand levels. Revenue increased from ₹449,926 in April to ₹532,979 in May, representing an 18.5% growth during peak demand. However, revenue declined sharply to ₹148,334 in June, showing a significant

72.2% decrease compared to May. In July, revenue increased slightly to ₹160,979, reflecting only an 8.5% recovery. The difference between peak and lowest revenue was ₹384,645, indicating high financial volatility caused by seasonal demand variation. The similarity between order volume and revenue trends confirms that business performance is directly influenced by demand fluctuations and lack of demand-based operational planning.

Key Findings:

- **Peak Demand and Revenue in May** -Order volume reached 393 orders and revenue increased to ₹532,979, representing 14.9% and 18.5% growth respectively, indicating maximum operational workload and financial performance.
- **Sharp Demand and Revenue Decline** -Orders dropped from 393 in May to 187 in June (52.4% decrease), while revenue declined from ₹532,979 to ₹148,334 (72.2% decrease), confirming strong seasonal fluctuation.
- **Limited Recovery in July** -Orders increased slightly to 194 (3.7% growth) and revenue increased to ₹160,979 (8.5% growth), indicating continued low demand and weak financial recovery.
- **Workload and Revenue Concentration** : April-May together contributed 735 orders (65.8%) and majority of revenue, whereas June-July accounted for only 381 orders (34.2%), confirming seasonal workload concentration.
- **Demand Driven Business Performance** : Revenue trend closely follows order volume, and the ₹384,645 gap between peak and lowest revenue highlights financial instability due to absence of demand-based planning.

3.3.2 Service-wise Seasonal Demand Comparison

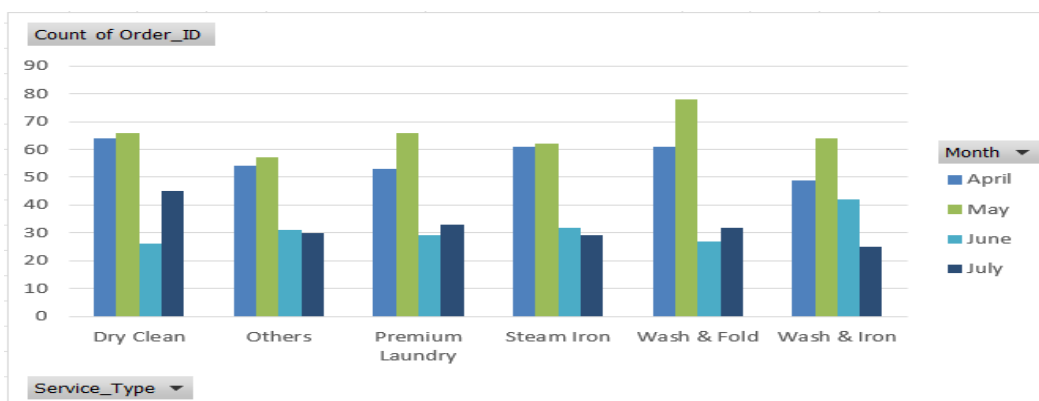


Figure 7: Service-wise Seasonal Demand Comparison -All Service Categories (April-July 2025)

The chart compares service- wise order distribution across months. Total orders peaked in May (393) and declined to lowest in June (187). Wash & Fold recorded highest demand in May (78 orders), followed by Dry Clean (66) and Premium Laundry (66). April-May together recorded 735 orders, while June-July recorded 381 orders, indicating a 48.2% decline. This confirms strong seasonal demand variation affecting operational efficiency.

Key Findings:

- **Peak Demand Month** - May recorded highest total orders of 393, indicating maximum operational workload.
- **Service- wise Demand Spike** - Wash & Fold recorded highest demand (78 orders) followed by Dry Clean and Premium Laundry (66 each).
- **Lowest Demand in June** - Total orders dropped to 187, representing significant seasonal decline.
- **Seasonal Demand Gap** - April-May recorded 735 orders vs 381 in June-July, showing 48.2% drop.
- **Workload Variation** - Demand fluctuation across months created operational strain during peak periods.

4. INTERPRETATIONS OF RESULTS AND RECOMMENDATIONS

4.1 Problem Statement: Long Turnaround Time & Service Delays

Interpretation

- **Low On Time Delivery Performance** : Out of 835 completed orders, only 317 orders were delivered on time, representing 37.9% of total completed orders. The remaining 518 orders (62.1%) experienced delays, indicating that the majority of services were not completed within expected timelines. This low on-time delivery rate reflects significant turnaround inefficiency and affects service reliability and customer satisfaction.
- **Recurring Moderate Delay Pattern** : The most frequent delay range observed was between 2-3 days, with 101 orders delayed by 2 days and 100 orders delayed by 3 days, together accounting for 201 delayed orders. These recurring moderate delays indicate systematic operational bottlenecks rather than isolated incidents, suggesting inefficiencies in processing

workflow and staff allocation.

- **Severe Delay Backlog During Peak Demand** : Delays of 4-5 days accounted for 202 orders, representing 24.2% of completed orders. These severe delays highlight backlog accumulation during high workload periods. The average delay of 1.83 days further confirms that service delivery consistently exceeded expected turnaround time.
- **Uneven Staff Workload Distribution** :Staff handled between 166 and 204 orders, indicating significant workload imbalance. Staff members assigned higher order volumes recorded greater delay frequency, while underutilized staff handled fewer orders despite available capacity. This uneven allocation contributed directly to increased turnaround time.
- **Direct Workload Delay Relationship** :During peak demand months, overloaded staff experienced processing backlog while underutilized staff operated below capacity. This confirms that operational capacity was not proportionally adjusted with demand variation, resulting in recurring service delays and reduced operational efficiency.

Recommendations

- **Implement Turnaround Time Monitoring** : Track delay days daily and flag orders exceeding 2-day delay for priority processing to prevent backlog accumulation.
- **Balanced Workload Allocation** : Distribute orders evenly across staff members to reduce workload imbalance and improve service efficiency.
- **Increase Processing Capacity During Peak Months** : Add temporary staff during high-demand periods to manage workload effectively.
- **Introduce Shift-Based Processing** : Add additional processing shifts during peak demand to improve turnaround time.
- **Dynamic Order Reassignment** : Transfer orders from overloaded staff to underutilized staff to reduce delay frequency.

Implementation Impact

Balancing staff workload and improving turnaround monitoring will reduce moderate delays and prevent backlog accumulation. Increasing processing capacity during peak demand will improve on-time delivery rate, enhance service reliability, and reduce customer dissatisfaction. These improvements will optimize operational efficiency and ensure consistent service performance.

4.2 Problem Statement: High Order Cancellations & Operational Disruptions

Interpretation

- **High Overall Cancellation Rate** : Out of 1,116 total orders, 281 orders were cancelled, resulting in an overall cancellation rate of 25.18%. This indicates that approximately one out of every four orders was not successfully completed. Such a high cancellation percentage reflects operational inefficiency and leads to revenue loss, reduced customer satisfaction, and unstable service reliability.
- **Peak Month Cancellations Indicate Capacity Constraint** : Cancellation rates were highest during peak demand months, with April recording 100 cancellations (29.24%) and May recording 113 cancellations (28.75%). In contrast, June recorded 38 cancellations (20.32%) and July recorded 30 cancellations (15.46%). The decline during low-demand months confirms that operational capacity remains fixed while demand fluctuates, leading to service failures during peak workload periods.
- **Late Pickup as Dominant Cancellation Cause** : Cancellation reason analysis shows that 178 out of 281 cancellations (63.3%) occurred due to late pickup. This indicates inefficiencies in pickup scheduling, route planning, and order handling. Delays at the pickup stage prevent orders from entering processing, leading to direct service failure.
- **Workforce and Resource Limitations** : Staff shortage accounted for 62 cancellations (22.1%), while inventory shortage accounted for 41 cancellations (14.6%). These figures indicate that operational resources are insufficient during high demand periods. Lack of workforce and resource planning contributes to service disruptions and increases cancellation probability.
- **Demand Driven Operational Disruptions** : Cancellation rates increased during peak demand months and decreased during low-demand periods, confirming that operational capacity does not scale with demand. This mismatch between order volume and service capability leads to pickup delays, staff overload, and increased cancellations.

Recommendations

- **Implement Capacity Based Booking** : Limit daily order intake based on processing and pickup capacity to prevent operational overload during peak demand.

- **Optimize Pickup Scheduling** : Introduce area-wise routing and fixed pickup time slots to reduce late pickups and improve order handling.
- **Increase Pickup Workforce During Peak Months** : Add temporary pickup executives during high-demand months to manage increased workload.
- **Maintain Inventory Buffer** : Keep additional consumables and packaging materials ready before peak demand periods to reduce cancellations.
- **Introduce Order Priority System** : Prioritize urgent and high-value orders to improve successful completion rate and reduce cancellations.

Implementation Impact

Improving pickup scheduling and increasing workforce capacity during peak demand will reduce cancellation probability and improve order completion rate. Capacity-based booking and inventory planning will prevent operational overload, stabilize order fulfillment, and enhance customer satisfaction while reducing revenue loss.

4.3 Problem Statement: Seasonal Demand Fluctuation

Interpretation

- **Strong Seasonal Demand Variation in Order Volume** : The monthly order trend shows significant fluctuation, with orders increasing from 342 in April to 393 in May, representing a 14.9% growth and peak operational workload. However, demand declined sharply to 187 orders in June, showing a 52.4% drop compared to May. In July, demand slightly recovered to 194 orders, representing only a 3.7% increase. This uneven distribution confirms strong seasonal demand variation and inconsistent workload across months.
- **Revenue Directly Dependent on Demand Pattern** : Revenue followed the same seasonal trend as order volume, increasing from ₹449,926 in April to ₹532,979 in May (18.5% growth), followed by a sharp decline to ₹148,334 in June, representing a 72.2% drop. July recorded a slight recovery to ₹160,979 (8.5% increase). The ₹384,645 gap between peak and lowest revenue highlights financial instability caused by seasonal demand fluctuations.
- **Peak Month Workload Concentration** : April and May together recorded 735 orders, contributing 65.8% of total demand, while June and July recorded only 381 orders (34.2%).

This concentration indicates that operational workload is heavily skewed toward peak months, leading to staff overload, processing delays, and increased cancellation probability.

- **Service wise Seasonal Demand Variation :** Service wise comparison shows that Wash & Fold recorded highest demand during peak months, reaching 78 orders in May, followed by Dry Clean and Premium Laundry services. In contrast, all service categories recorded significant decline in June, confirming that seasonal variation affects multiple service types simultaneously.

- **Operational Capacity Mismatch with Demand :** While demand fluctuated significantly between months, staffing and operational capacity remained constant. Peak months experienced higher delays and cancellations due to overload, whereas off-season months experienced smoother operations. This mismatch confirms lack of demand-based planning and proactive resource allocation.

Recommendations

- **Implement Seasonal Workforce Planning :** Increase staffing levels during peak demand months (April-May) by 15-20% to handle increased order volume efficiently.
- **Introduce Demand-Based Scheduling :** Adjust pickup slots, processing capacity, and delivery timelines based on expected monthly demand patterns.
- **Plan Capacity Expansion Before Peak Months :** Deploy temporary staff, additional machines, and extended shifts prior to high-demand periods.
- **Introduce Seasonal Pricing and Promotions :** Offer off season discounts during June-July to stabilize demand and reduce revenue volatility.
- **Implement Demand Monitoring Dashboard :** Track service wise demand trends and trigger operational scaling when monthly orders exceed threshold levels.

Implementation Impact

Seasonal demand based planning will reduce operational overload during peak months and improve resource utilization during low-demand periods. Aligning staffing, scheduling, and processing capacity with demand variation will stabilize revenue, reduce service delays, minimize cancellations, and improve overall operational efficiency.